Ergonomics

ISSN: 0014-0139 (Print) 1366-5847 (Online) Journal homepage: www.tandfonline.com/journals/terg20

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To cite this article: Peter J Keir & Christopher W MacDonell (2004) Muscle activity during patient transfers: a preliminary study on the influence of lift assists and experience, Ergonomics, 47:3, 296-306, DOI: 10.1080/0014013032000157922

To link to this article: https://doi.org/10.1080/0014013032000157922

Published online: 20 Feb 2007.

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ERGONOMICS, 26 FEBRUARY, 2004, VOL. 47, NO. 3, 296–306

Taylor & Francis healthsciences

Muscle activity during patient transfers: a preliminary study on the influence of lift assists and experience

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Keywords: EMG; Patient transfer; Lifting.

The purpose of this study was to examine muscle activity patterns during patient handling during manual transfers, and transfers using floor and ceiling lifts. EMG patterns during transfers from bed to wheelchair and wheelchair to bed as well as patient repositioning in novices and experienced participants were examined. Surface EMG was recorded from the upper and lower erector spinae, latissimus dorsi and trapezius muscles bilaterally. Overall, normalized mean and peak muscle activity were lowest using the ceiling lift, increasing with the floor lift, which were lower than manual transfers (novices: all $p < 0.01$). Experienced patient handlers demonstrated approximately two times greater trapezius and latissimus dorsi activity than novices, combined with lower mean erector spinae activity ($p < 0.05$, for most tasks). Integrated EMG for all muscles was directly proportional to the transfer time and was lowest during the manual transfer followed by the ceiling lift, with the floor lift being highest. The difference between the muscle activity patterns between the experienced and novice patient handlers may suggest a learned behaviour to protect the spine by distributing load to the shoulder. Further examination of the muscle activation patterns differences between experience levels could improve training techniques to develop better patient handling strategies.

1. Introduction

The health care industry has long recognized that low back pain (LBP) is a major concern with respect to patient handling (Stubbs *et al.* 1983), with the risk of LBP increasing with the frequency of patient handling (Jensen 1990). Nursing aides, who handle patients more frequently than nurses, are a prime example, reporting more low back pain, injuries and compensation claims than nurses (Videman *et al.* 1984, Pheasant and Stubbs 1992). Findings such as these have prompted examination of various aspects of patient handling, including, the loads acting on the lumbar spine (Garg *et al.* 1991a, b, 1992, Marras *et al.* 1999, Daynard *et al.* 2001), teaching and training methods (Owen *et al.* 1999, Johnsson *et al.* 2002), as well as mechanical interventions (Garg and Owen 1992, Brophy *et al.* 2001, Yassi *et al.* 2001).

Over the past two decades, biomechanists have demonstrated that compression and shear loads on the lumbar spine during patient handling are high (Gagnon *et al.* 1987, Garg *et al.* 1991a, b, 1992, Marras *et al.* 1999, Daynard *et al.* 2001). These

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Ergonomics ISSN 0014-0139 print/ISSN 1366-5847 online © 2004 Taylor & Francis Ltd
http://www.tandf.co.uk/journals
DOI: 10.1080/0014013032000157922

forces often exceed the Action Limit (AL, 3400 N) and Maximum Permissible Limit (MPL, 6400 N) for lumbar spine compression set forth by the National Institute for Occupational Safety and Health (NIOSH, Waters *et al.* 1993). Single person and tandem manual transfers have compressive loads in the lumbar spine in excess of both NIOSH limits (Winkelmolen *et al.* 1994, Marras *et al.* 1999). For this reason, amongst others, single person manual transfers and lifts are no longer advocated but are still often performed by many workers (Owen *et al.* 1999). Gagnon *et al.* (1987) demonstrated that these high forces are not limited to lifting or transferring patients, but the act of simply turning a patient in bed was associated with mean L5/S1 disc compression forces up to 3526 N when the patient was turned rapidly with the knees supported on a low bed.

Historically, lumbar joint moments (about the L4/L5 or L5/S1 intervertebral disc), shear and compression forces have been presented as mean (average) or peak values and compared to tolerance data (Gagnon *et al.* 1987, Garg *et al.* 1991a, b, 1992, Marras *et al.* 1999, Zhuang *et al.* 1999, Daynard *et al.* 2001). Examination of spinal loading in a large cohort study found that cumulative lumbar compression over an entire shift was associated with injury in automotive workers (Norman *et al.* 1998). These same techniques were used in an investigation of the compressive loads accumulated in nursing personnel over the course of a shift and indicated that the cumulative loading on the lumbar spine was greater in some tasks when using assistive devices than when performed manually, due to increased duration of patient transfers with the mechanical assists (Daynard *et al.* 2001).

Based on their EMG data, Gagnon *et al.* (1987) queried whether a strategy of minimizing spinal compression or minimizing shoulder moments should be promoted; acknowledging that transferring the problem from the low back to another joint must be considered in any recommendations. The inclusion of shoulder activity by Gagnon *et al.* (1987) is notable as most studies have focused solely on the lumbar spine. Although advanced spinal models (Marras *et al.* 1999) require extensive EMG data for input, reports of actual muscle activity levels during patient transferring tasks are limited, especially for muscles other than the back extensors. Subjective measures, such as 'stress levels' (often called ratings of perceived exertion), have been used to indicate worker perceptions of loading on the low back, shoulder, upper back and whole body (Garg *et al.* 1991a). These measures, while providing important information, are limited as they include subjective factors other than muscle and joint loading.

Lifting and transfer aids, including specialized transfer sheets, slings, belts, as well as various floor and ceiling lift devices, have been developed to lower the demands on patient handlers. Beyond the mechanics of using these aids, the perceptions of the workers (Garg *et al.* 1991b, Garg and Owen 1992, Caboor *et al.* 2000, Zhuang *et al.* 2000) and the patients being transferred (Garg *et al.* 1991b, Zhuang *et al.* 2000) have shown that some devices are more accepted than others. In recent years, research into the musculoskeletal loads experienced by patient handlers has increased and the concept of 'zero lift' in hospitals has been introduced to reduce the risk of injury (e.g. Rockefeller *et al.* 2000). Mechanical assists relieve some of the physical burden of the patient transfer but often require an increased duration, which may lead to increased cumulative loading in spite of a reduction in peak spinal loads (Daynard *et al.* 2001). This may have implications for the risk of injury as well as guidelines for using mechanical assists (Daynard *et al.* 2001).

While analyses of lumbar compression and shear are very meaningful to biomechanists and useful in the context setting guidelines by agencies such as NIOSH, these terms often have limited meaning to the lay public and workers who are most affected. EMG analysis of muscles in addition to the erector spinae during patient transfers using different assist devices may improve the understanding of overall muscular loading during patient transfers. The purpose of the present study was to investigate the response of the trunk and shoulder musculature in novice and experienced patient handlers during manual transfers and transfers using a floor lift and a ceiling lift.

2. Methods

2.1. *Participants*

Seven volunteers participated in the study. Four had little or no experience in patient handling and were classified as novices (all male). A fifth novice participated in the study but was excluded from the analysis due to technical difficulties. Three volunteers (2 male, 1 female) were experienced with all transfer methods used in the study. Novices had a mean age of 25.0 ± 2.2 years, were 180.3 ± 12.4 cm tall and weighed 81.5 ± 4.0 kg. The experienced group were 35.0 ± 8.0 years of age, 175.7 ± 15.7 cm tall and weighed 86.7 ± 17.2 kg. The experienced group consisted of individuals who were involved with training and instruction, but were not employed as health care professionals. In a brief session prior to data collection, the novice patient handlers were trained in proper transfer techniques. All participants were informed of the study protocol and signed an informed consent form. The protocol was approved by the University Human Participants Review Committee.

2.2. *Protocol*

A compliant partial weight bearing 92 kg 'patient' was transferred from bed-to-wheelchair and wheelchair-to-bed under three conditions. The three transfer conditions were as follows: (1) single person manual ('hug' technique); (2) using a mobile pneumatic (hand pump) floor lift, and (3) using an electric ceiling lift mounted on a moveable gantry (trans*active*™, Waverley Glen Systems Ltd, Toronto, ON, Canada). Each transfer was completed twice. Both the floor and ceiling lifts made use of a transfer sling. Adjusting the sling prior to each lift was not included in the analysis. In addition to the transfers, repositioning the patient in the bed was also examined. The reposition consisted of moving a patient 'up' in the bed, or, shifting them towards the head of the bed. The bed remained horizontal at a constant height for these trials.

Surface EMG was used to examine muscle activity during the tasks. Bipolar surface electrodes were affixed over the muscle bellies of four left/right muscle pairs after shaving and cleansing of the skin with alcohol. The muscles examined were the right and left upper erector spinae (RUES, LUES), the lower erector spinae (RLES, LLES), the ascending portion of the trapezius (RTRP, LTRP), and the latissimus dorsi (RLAT, LLAT). Erector spinae electrodes were located at the T9 level for the UES and at the L3 level for the LES. EMG signals were pre-amplified $100\times$ near the source and band-pass filtered from $10-500$ Hz. Root-mean-square (RMS) EMG was collected using a 220 ms window and a sampling rate of 120 Hz (to synchronize with video). The gain on each EMG channel was adjusted to maximize the signal amplitude. EMG was normalized to

100% MVC based on a series of trials to elicit maximal contractions. The calibration trials included prone resisted back extension, resisted shoulder elevation and retraction, as well as voluntary isometric maximal efforts (cf McGill 1991).

All trials were recorded with video synchronized with the EMG data using a light emitting diode (LED) visible in the field of view to identify the start of the data collection. Data analysis included time to completion for each transfer, mean (average) EMG (AEMG) and the Amplitude Probability Distribution Function (APDF) of the RMS EMG profile. The tenth, fiftieth (median) and ninetieth (peak) percentiles of the APDF were analysed. Cumulative loading for each muscle was determined using the integral of the RMS EMG—time curve. Data were analysed using repeated measures ANOVA with *a priori* contrasts (STATISTICA, StatSoft, Inc., Tulsa, OK, USA). Unless an F-statistic is presented, all p-values reflect those determined by planned comparisons.

3. Results

On all measures, the results for bed-to-wheelchair and wheelchair-to-bed transfers were very similar. While all data are included in the figures, the results and discussion will focus on the transfers from bed-to-wheelchair, as discussion of both transfer directions would be redundant.

3.1. *Mean muscle activity (AEMG)*

The conditions of patient transfer had a significant effect on mean muscle activity (AEMG, figure 1). Experience significantly altered muscle activity ($F_{7,28} = 3.3227$,

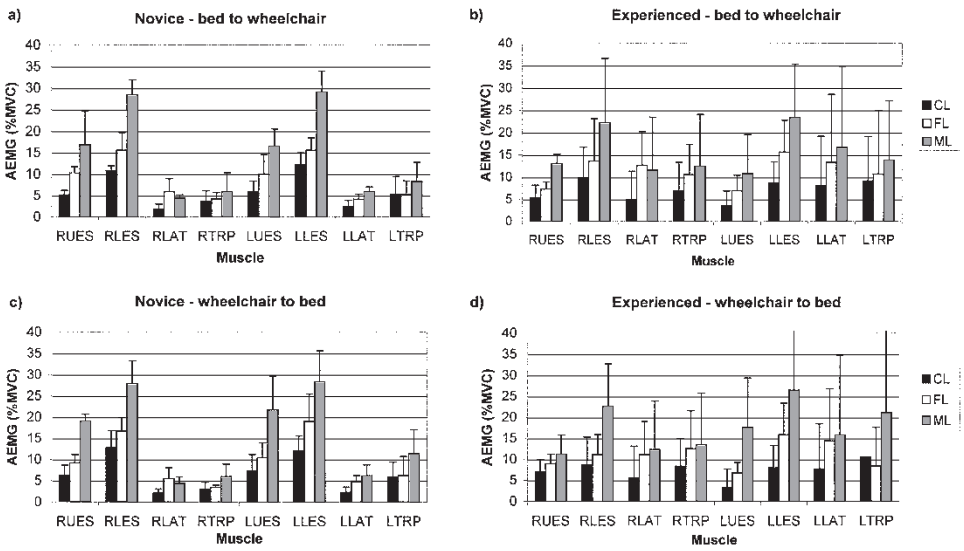

Figure 1. Mean EMG (plus standard deviation) during transfer tasks. (a) Novices, bed to wheelchair, (b) experienced, bed to wheelchair, (c) novices, wheelchair to bed, (d) experienced, wheelchair to bed. Novices (a, c) used slightly more erector spinae activity and less TRP and LAT activity than the experienced patient handlers (b, d). Legend abbreviations: CL = ceiling lift, FL = floor lift, and ML = manual transfer. Note: statistics and values in text refer to panels (a) and (b).

$p = 0.00106$). In novice patient handlers, the ceiling lift required significantly less muscle activation (pooled across all muscles) than the floor assist ($p < 0.008$), which was lower than the manual transfer ($p < 0.011$). Similarly, the experienced patient handlers required significantly less muscle activation when using the ceiling lift than either of the other two transfer conditions (floor, $p < 0.006$; manual, $p < 0.01$), while the floor lift had lower activity than the manual lift ($p < 0.05$). The type of lift also significantly affected muscle activity ($F_{14,56} = 3.6346$, $p = 0.00028$). Comparing the ceiling vs. the floor lift, and pooling across experience levels, RUES, RLAT, and RTRP were all significantly lower with the ceiling lift ($p < 0.01$). LUES, RLES and LLAT were also lower with the ceiling lift ($p < 0.05$).

In novices, highest mean muscle activities were found in the lower erector spinae (mean of right and left) at 11.5, 15.6 and 28.8% MVC for the ceiling, floor and manual lifts, respectively (figure 1a). The upper erector spinae (RUES and LUES) required slightly less mean activity at 5.6, 9.8 and 16.8% MVC for the ceiling, floor and manual lifts, respectively. In most trials, novice mean latissimus dorsi and trapezius activation levels were below 5–6% MVC, the exception being the LTRP in the manual transfer (figures 1a,b). The lowest activities were generally found with the ceiling lift and the highest with the manual transfer.

In the experienced group, the highest mean activity levels were found in the lower erector spinae and were similar to or slightly lower than novices, with the manual lift showing the greatest difference at about 5% less than novices. The upper erector spinae were also slightly lower than the novices but the trapezius and latissimus dorsi muscles in the experienced handlers were higher than the novices (e.g. RTRP, $p < 0.05$) (figures 1a and b).

3.2. *Peak muscle activity*

Peak muscle activity, defined as the ninetieth percentile of the APDF, exhibited a similar trend to the mean EMG but with larger amplitudes (figure 2). Experience had a marginally significant effect on muscle activity ($F_{7,28} = 3.0179$, $p = 0.0171$). Transfer condition had a significant effect on peak muscle activity in the novice patient handlers with the ceiling lift being lower than the floor lift which was significantly lower than the manual transfer (all $p < 0.005$). The same effect was present with the experienced group, with the exception that the difference between the ceiling and the floor was not as distinct ($p < 0.035$). When the experience levels were pooled, overall activation differences between the transfers were significant ($p < 0.005$). The differences in overall activation between the ceiling and floor lift were due to differences in RUES, both LES, and both LAT ($p < 0.05$).

As seen in figure 2, highest peak activity for novices was again found in the LES (both right and left) at 25.2, 29.2, and 43.6% MVC for ceiling, floor, and manual transfers, respectively. The upper erector spinae were next highest, again with the trapezius and latissimus dorsi having the lowest activity. For the novice group, the LAT muscles generally remained at or below 10% MVC and TRP ranged from less than 6% with the ceiling lift to over 24% (LTRP) during the manual transfer (figure 2c). Experienced patient handlers demonstrated slightly lower LES activity levels but the profile across all muscles appeared to be flat, indicating that the LAT and TRP peak EMG were similar to the erectors (figure 2b). As with AEMG, the LAT and TRP peak values of the experienced group were much higher than the novice group (e.g. $p < 0.05$ for RTRP).

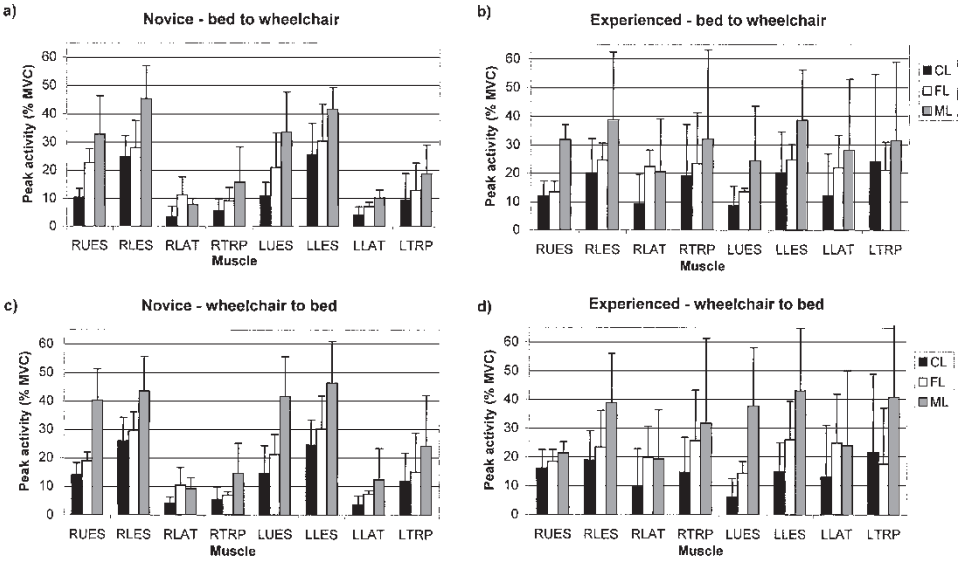

Figure 2. Peak EMG (90th percentile) data from the amplitude probability distribution function (APDF) for each condition. Presented are the 90th (0.9) percentile activation levels (plus standard deviations) for each muscle. (a) Novices, bed to wheelchair, (b) experienced, bed to wheelchair, (c) novices, wheelchair to bed, (d) experienced, wheelchair to bed. Legend abbreviations: CL = ceiling lift, FL = floor lift, and ML = manual transfer. Note: statistics and values in text refer to panels (a) and (b).

3.3. *Integrated response*

Both transfer time and integrated EMG response were greatest for the floor lift (table 1). This is due to a combination of high activation and the longest transfer time. Despite novices requiring over 50% more time with the floor lift (novices 144.8 s, experienced 91.9 s), IEMG was greater for the experienced group for right and left LAT and TRP muscles, but less for the upper and lower erectors on both sides (LUES, RUES, LLES, RLES). Similar results were found for the ceiling lift but the difference for the UES was not as prominent. Despite eliciting the lowest muscle activity, the duration of the ceiling lift aided transfers resulted in larger integrated values than the manual transfer, which required the greatest muscle activity but only a fraction of the time, resulting in low integrated values. Based on the preliminary nature of this study and the large variances in the integrated muscle activations, the authors opted to present only descriptive statistics, thus no statistical significance is implied from the above comparisons as well as those in table 1.

3.4. *Repositioning*

The peak EMG values (90th percentile APDF) for the two-person repositioning task are found in table 2. As can be seen from table 2, the differences in peak activity between the experience levels are not as pronounced as in the transfer tasks.

4. Discussion

In this laboratory study, muscle activity was lowest during patient transfers using the ceiling lift, increasing with the floor lift, and was highest during single person manual

Table 1. Integrated EMG response. Bed to wheelchair transfer (units = %MVC.s). Bold typeface indicates when the novices had greater IEMG and italicized typeface indicates experienced IEMG was greater than novices. Note: comparisons are descriptive only and do not imply significance.

	Time (s)	RUES	RLES	RLAT	RTRP	LUES	LLES	LLAT	LTRP
Ceiling									
Nov	91.7	457	**1098**	*180*	*304*	**567**	**1126**	*198*	*532*
	(13.4)	(189)	**(199)**	*(126)*	*(215)*	**(255)**	**(168)**	*(136)*	*(433)*
Exp	78.5	462	**781**	*597*	*623*	**393**	**752**	*628*	*727*
	(11.7)	(221)	**(530)**	*(473)*	*(582)*	**(293)**	**(460)**	*(847)*	*(709)*
Floor									
Nov	144.8	**1464**	**2158**	*788*	*594*	**1493**	**2314**	*602*	*694*
	(30.5)	**(306)**	**(184)**	*(319)*	*(179)*	**(841)**	**(660)**	*(198)*	*(318)*
Exp	91.9	**659**	**1249**	*1150*	*958*	**632**	**1435**	*1203*	*966*
	(2.7)	**(141)**	**(842)**	*(663)*	*(610)*	**(312)**	**(641)**	*(1371)*	*(1285)*
Manual									
Nov	17.4	298	495	*78*	*99*	288	503	*104*	*141*
	(1.0)	(148)	(41)	*(14)*	*(68)*	(65)	(60)	*(25)*	*(71)*
Exp	22.7	306	453	*243*	*326*	217	485	*336*	*299*
	(6.4)	(58)	(210)	*(206)*	*(377)*	(138)	(149)	*(307)*	*(246)*

Table 2. Peak (90th percentile) EMG values in percent MVC from APDF for two person repositioning for both right and left repositions.

	RUES	RLES	RLAT	RTRP	LUES	LLES	LLAT	LTRP
Right								
Nov	42.3	51.7	8.8	20.7	27.5	38.2	12.1	21.8
	(19.3)	(19.3)	(3.6)	(4.0)	(17.8)	(16.4)	(5.5)	(14.2)
Exp	19.8	44.0	17.6	41.5	19.0	19.6	23.8	29.8
	(11.6)	(24.3)	(1.4)	(35.2)	(5.1)	(14.4)	(4.6)	(36.9)
Left								
Nov	26.6	46.3	20.8	21.7	20.0	45.3	19.3	15.8
	(37.8)	(17.8)	(11.9)	(14.5)	(5.3)	(12.0)	(14.8)	(7.1)
Exp	16.5	40.2	21.5	25.0	24.3	24.4	28.6	34.7
	(4.9)	(26.8)	(9.4)	(24.5)	(16.9)	(16.3)	(11.6)	(38.0)

transfers. Although it required the highest activation levels, the manual lift required the shortest time to complete at only a fraction of the other two devices; the ceiling lift taking much less time than the floor lift, especially in novices. The combination of these factors was examined through the integrated EMG response, which was smallest for the manual lift, followed by the ceiling lift and highest when using the floor lift. This was generally true for all muscles but was especially noticeable in the back extensor muscles. Increased use of the latissimus dorsi and trapezius muscles by the experienced patient handlers was also found, which may be a protective strategy.

With lower muscle activity, there is an obvious benefit to using the mechanical lifts, especially the ceiling lift. This indicates that peak loading of muscles, and subsequent spinal compression, are lower for the ceiling lift, decreasing the acute risk from overload. However, the integrated response was found to be lowest for the manual lift, as previously found for cumulative lumbar compression, also attributed to shorter transfer time (Daynard *et al.* 2001). This appears to cloud the issue of

implementing 'zero-lift' policies in hospitals and long-term care facilities as suggested by Daynard *et al.* (2001). Whilst the contributions of cumulative loading to spinal injury should not be dismissed, the link between peak and mean spinal loading (compression and/or moments) is particularly well documented (e.g. Marras *et al.* 1993) and should not likely be superseded by cumulative loading until further support has been presented. The risk associated with LBP is likely greater when peak and cumulative loading are both high, as they appear to be independent risk factors (Norman *et al.* 1998). Additionally, increased lumbar compression and subsequent reporting of LBP has been found in 'new', 'younger', and untrained or non-compliant workers (Videman *et al.* 1984, Daynard *et al.* 2001), indicating that safe patient transfer technique may be learned through experience. Interestingly, it appears that training alone may be insufficient to reduce injuries, suggesting that the ultimate goal should be to minimize the forces on the body associated with patient transfers (Pheasant and Stubbs 1992, Yassi *et al.* 1995, Daynard *et al.* 2001). Although there has been some evidence to the contrary, these factors and the knowledge that injury rates increase with lifting frequency (Jensen 1990), indicate that not using mechanical devices may put workers at risk.

The trade-off between muscle activity level and transfer time is particularly noteworthy. For example, the mean manual transfer time was under 20 s but required lower erector spinae (LES) activity of almost 30% MVC (AEMG) while the ceiling lift had a mean of just under 100 s with AEMG levels of less than 13% MVC. The floor lift required lower activation than the manual transfer, but took longer than the ceiling lift to complete (over 120 s). This trade-off must be considered to be a contributing factor in the reported reluctance of health care professionals to use lift assists, even when provided (e.g. Daynard *et al.* 2001). The integrated EMG response was included to allow for a relevant comparison to cumulative spinal loading which has been shown to be an important factor in determining risk of LBP (Norman *et al.* 1998). EMG activity of the lower erector spinae has been demonstrated to be a good predictor of lumbar compression (Potvin *et al.* 1990) and been promoted as a method for determining lumbar compression in epidemiology studies in ergonomics (Wells *et al.* 1997). However, the generalizability of the IEMG measures in this study *may be* limited as large variances were present, likely due to the short duration of the tests, small sample size, and the varied skill levels of our volunteers.

This study focused on muscle activity of the back and shoulder girdle muscles and, although the sample size was small, the comparison between experienced patient handlers and novices proved interesting. Both mean and peak trapezius and latissimus dorsi activity were twice as high for the experienced vs. the novice handlers. This was true for all transfer methods and not just the manual transfer. Single person manual transfers were included in the study even though they are no longer advocated (Owen *et al.* 1999) to provide an estimate of the maximum forces that might be incurred by a patient handler under any circumstance. It appeared that the experienced patient handlers opted for a strategy that reduced erector spinae activity, and increased the use of the shoulder girdle muscles as previously suggested (Gagnon *et al.* 1987). This would act to reduce the compressive forces on the lumbar spine but may increase the risk of shoulder injuries. This was further examined by using an approximation of L4/L5 compression based on lower erector spinae EMG (Potvin *et al.* 1990) on two of the experienced handlers and found relatively low compression (well below 3400 N). This may indicate that these two individuals may have consciously reduced spinal compression by reducing erector spinae activity.

Review of the video indicated that these individuals maintained very erect spine when performing the manual transfers. While not solely related to patient transfers, Knibbe and Friele (1996) found that sick leave incidence due to musculoskeletal disorders other than LBP exceeded those due to LBP alone. Whether these data are indicative of a learned behaviour to protect the lumbar spine remains to be determined.

Repositioning using a draw sheet sling indicated little difference between the experienced patient handlers and the novices, in contrast to the transfers. This may be due to the nature of the experienced group who do not reposition patients on a regular basis or it may simply indicate that a two-person reposition with a sling is a fairly constrained task. With the exception of one participant, the person assisting with the reposition was constant, thus some degree of consistency can be assumed. There also appeared to be no directional effect of the reposition on muscle activity, with right and left muscle activation levels being similar whether the reposition was to the left or the right.

There are several limitations to the present study. First, our experienced patient handlers were not health care practitioners but professionals who instruct individuals on proper transfer techniques especially with the mechanical devices. Patient weight is an important consideration. Previous studies have ranged from 50–100 kg 'patients', sometimes including light and heavy patients (Marras *et al.* 1999). The choice of a 92 kg 'patient' in this study represents a heavier patient which would be more of a concern for the patient handler than a light patient. In addition to better reflecting the patient population of concern, differences in tissue loading may be better elucidated with heavier patients (Zhuang *et al.* 1999). Finally, we did not include the full act of placing and using the sling, only the final act of hooking it to the lift assists. The end effect of this decision is likely a reduction in transfer time and cumulative muscle loading but not the peak values. With respect to sling use, transfers with the floor and ceiling lifts are directly comparable.

This study, while preliminary in nature, has provided important additions to the recent studies on peak and cumulative loading of the *lumbar spine* and shoulder girdle muscles during the transfer of patients with lift assists. While cumulative loading may provide additional insight (Norman *et al.* 1998), it is likely that peak loading is of utmost concern when transferring patients (Daynard *et al.* 2001). By analysing mean, peak and cumulative EMG from the trapezius and latissimus dorsi muscles, in addition to the erector spinae, differences between lift assists and experience level were found. Further research into the external forces and muscle activity levels during the normal workday in a health care facility must be conducted to fully evaluate the specific tissues at risk in health care workers. To determine the muscle control strategies of patient handlers, further examination of contributing muscles, in addition to the erector spinae, during normal work conditions is required. Further examination of muscle activation pattern differences between experience levels could lead to improvements in training techniques and better patient handling strategies.

Acknowledgements

The authors wish to thank Jeremy Mogk for his assistance with data analysis. Lift assists, hospital equipment and financial assistance were provided by Waverley Glen Systems, Ltd. Additional funding was provided by Natural Sciences and Engineering Research Council grant # 217382-00 (PJK).

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